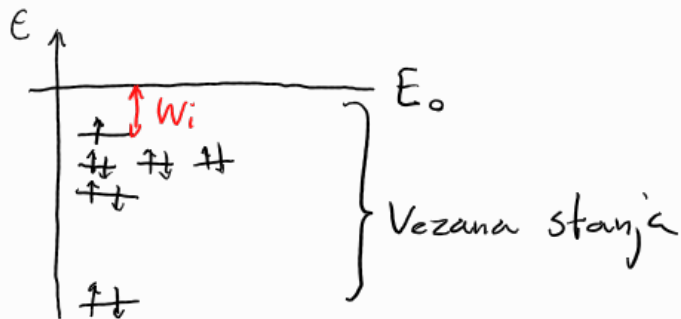


① Molekula NaCl

• $\text{Na} \rightarrow \text{Na}^+ + e^-$: potrebna je ionizacijska energija W_i

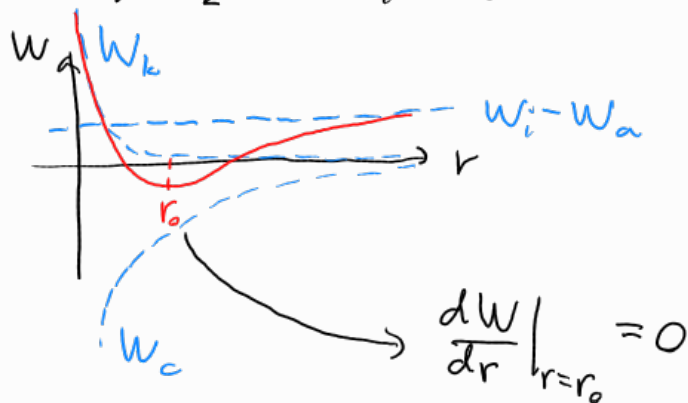


• $\text{Cl} + e^- \rightarrow \text{Cl}^-$: sprosti se elektronska afiniteta $-W_a$

• Coulombska interakcija + kvantni odboj (Pauli)

$$W_c = \frac{e^2}{4\pi\epsilon_0 r} \quad W_k = \frac{C}{r^\beta}, \beta > 1, \text{ dobljeno z eksperimentom}$$

$$\rightarrow W(r) = W_k + W_c + W_i - W_a$$



• posplošitev na kristal: W_p na par : $W(r) = W_c(r) + W_k(r) + W_i - W_a$

$$- \frac{e^2 Z \alpha_n}{4\pi\epsilon_0 r}$$

Madelungova konstanta
(vpliv okolice)

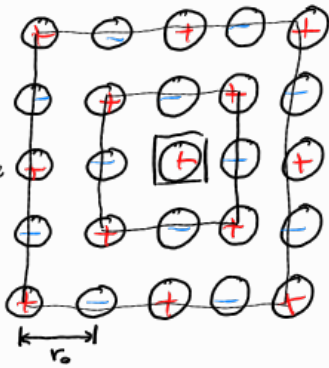
Z : naboj ; X^{z+}/X^{z-}

4.1 Zorko

2D kristal

$\alpha_n = ?$

(upoštevaj prispevke
prvih dveh kvadr.)



$$\left. \begin{array}{l} \text{red } + \rightarrow e = +e_0 \\ \text{blue } - \rightarrow e = -e_0 \end{array} \right\} \Rightarrow N_+ = N_- = N$$

seštevamo po kvadratih:
(območje od enega do
drugega kvadrata)

$$\alpha_n^{(1)} = \frac{4 \cdot (-1/2)}{r_0/2} + \frac{4(1/4)}{\sqrt{2} r_0/2} = -1,293$$

$$\alpha_n^{(2)} = \frac{4(-1/2)}{1} + \frac{4(3/4)}{\sqrt{2}} + \frac{4(1/2)}{2} + \frac{8(-1/2)}{\sqrt{5}} + \frac{4(1/4)}{2\sqrt{2}} = -0,314$$

$$\rightarrow \alpha_n \doteq \alpha_n^{(1)} + \alpha_n^{(2)} = -1,61$$

$$W_{\text{krist.}} = 2N \cdot \overbrace{5\text{eV}}^{W_0}$$

$$W_{\text{krist.}} = N \cdot W_{\text{par}}(r_0) = N(W_k + W_c + W_i - W_a)$$

$$W_i = 15\text{eV}$$

$$W_a = 10\text{eV}$$

$$W_k \propto r^{-12}$$

$$r_0 = ?$$

$$-2W_0 = W_k + W_c + W_i - W_a = \frac{k}{r_0^{12}} + \frac{e_0^2 \alpha_n}{4\pi \epsilon_0 r_0} + W_i - W_a$$

$$\left. \frac{dW}{dr} \right|_{r=r_0} = -\frac{12k}{r_0^{13}} - \frac{e_0^2 \alpha_n}{4\pi \epsilon_0 r_0^2} = 0 \Rightarrow k = -\frac{e_0^2 \alpha_n r_0^{11}}{48\pi \epsilon_0}$$

$$-2W_0 = -\frac{e_0^2 \alpha_n}{48\pi \epsilon_0 r_0} + \frac{e_0^2 \alpha_n}{4\pi \epsilon_0 r_0} + W_i - W_a = \frac{11e_0^2 \alpha_n}{48\pi \epsilon_0 r_0} + W_i - W_a$$

$$\rightarrow r_0 = \frac{11e_0^2 \alpha_n}{48\pi \epsilon_0 (-2W_0 - W_i + W_a)} = \underline{0,20\text{nm}}$$

4.2 Zorko
 N_{ACL}
 $r_0 = 0,281$
 $\alpha_n = 1,75$
 $W_L(r) = 9/7,7$

$k = ?$ (kako se W obnaša v okolici r_0 / kakšno bo nihanje ionov, če jih malo izmaknemo iz ravnl. lege)

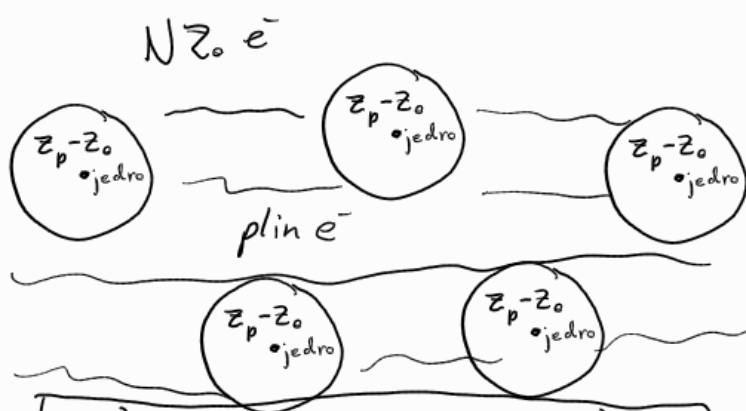
$$\left. \frac{dW}{dr} \right|_{r=r_0} = 0 \Rightarrow C = \frac{\alpha_n e_0^2 r_0^{6,7}}{4 \cdot 7,7 \pi \epsilon_0}$$

$$k = \frac{\partial^2 W}{\partial r^2} = \frac{(-7,7)(-8,7)C}{r_0^{9,7}} = \frac{\alpha_n e_0^2 (-1)(-2)}{4 \pi \epsilon_0 r_0^3} = \boxed{\frac{6,7 \alpha_n e_0^2}{4 \pi \epsilon_0 r_0^3}}$$

$$\omega = \sqrt{\frac{k}{m^*}} = \boxed{7,2 \cdot 10^{13} \text{ s}^{-1}} \sim \text{THz}$$

↑
reducirana masa

Kovine



$$E = B = 0 \Rightarrow \langle \vec{p} \rangle = 0$$

$E \neq 0$:

$$\boxed{\frac{d\langle \vec{p} \rangle}{dt} = -e_0 (\vec{E} + \vec{v} \times \vec{B}) - \frac{\langle \vec{p}(t) \rangle}{\tau}}$$

prevodnost

$$\langle \vec{v} \rangle = \frac{\langle \vec{p} \rangle}{m}, \quad \vec{j} = n e_0 \langle \vec{v} \rangle = n e_0 \frac{\langle \vec{p} \rangle}{m} = \sigma \vec{E}$$

$$E \neq 0, \quad \frac{d\langle \vec{p} \rangle}{dt} = 0$$

$$\langle \vec{p}(t) \rangle = -e_0 \tau \vec{E}$$

$$\rightarrow \boxed{\vec{j} = n e_0 \langle \vec{v} \rangle = - \frac{n e_0^2 \tau}{m} \vec{E}}$$

$\sigma \rightarrow$ prevodnost

5) $\vec{E} = \hat{e}_x E_0 \begin{cases} 1; t < 0 \\ e^{-\lambda t}; t \geq 0 \end{cases}$

Natana, $v(t=0) = v_0$

$e = ?$

$\frac{d\langle \vec{p} \rangle}{dt} = -e_0 \vec{E} - \frac{\langle \vec{p}(t) \rangle}{\tau} \rightarrow 1) t < 0: \frac{d\langle \vec{p} \rangle}{dt} = 0 \Rightarrow \langle \vec{p} \rangle = -\tau e_0 E_0 \hat{e}_x$

2) $t \geq 0$: i) $\langle \vec{p}_h \rangle$: $\frac{d\langle \vec{p} \rangle}{dt} = -\frac{\langle \vec{p} \rangle}{\tau}$

$\ln \langle \vec{p} \rangle = -t/\tau + c$

$\langle \vec{p} \rangle = D e^{-t/\tau}$

ii) $\langle \vec{p}_p \rangle$: $\vec{p} = D(t) e^{-t/\tau}$

$\dot{\vec{p}} = \dot{D}(t) e^{-t/\tau} - \frac{1}{\tau} D(t) e^{-t/\tau}$

$e^{-t/\tau} (\dot{D} - \frac{1}{\tau} D) = -e_0 E_0 \hat{e}_x e^{-\lambda t} - \frac{1}{\tau} D e^{-t/\tau}$

$\dot{D} = -e_0 E_0 \hat{e}_x e^{-t(\lambda - 1/\tau)}$

$D = e_0 E_0 \hat{e}_x \frac{1}{\lambda - 1/\tau} e^{-t(\lambda - 1/\tau)}$

$\Rightarrow \langle \vec{p} \rangle = D e^{-t/\tau} + \frac{e_0 E_0 \hat{e}_x}{\lambda - 1/\tau} e^{-t(\lambda - 1/\tau)} e^{-t/\tau} =$
 $= D e^{-t/\tau} + e_0 E_0 \hat{e}_x \frac{\tau}{\lambda \tau - 1} e^{-\lambda t}$

Robni pogoj: $\langle \vec{p}(0) \rangle = -\tau e_0 \hat{e}_x E_0 \rightarrow \langle \vec{p}(0) \rangle = D + e_0 E_0 \hat{e}_x \frac{\tau}{\lambda \tau - 1}$

$\rightarrow D = -e_0 E_0 \hat{e}_x \tau \left(1 + \frac{1}{\lambda \tau - 1} \right) = -e_0 E_0 \tau \hat{e}_x \left(\frac{\lambda \tau}{\lambda \tau - 1} \right)$

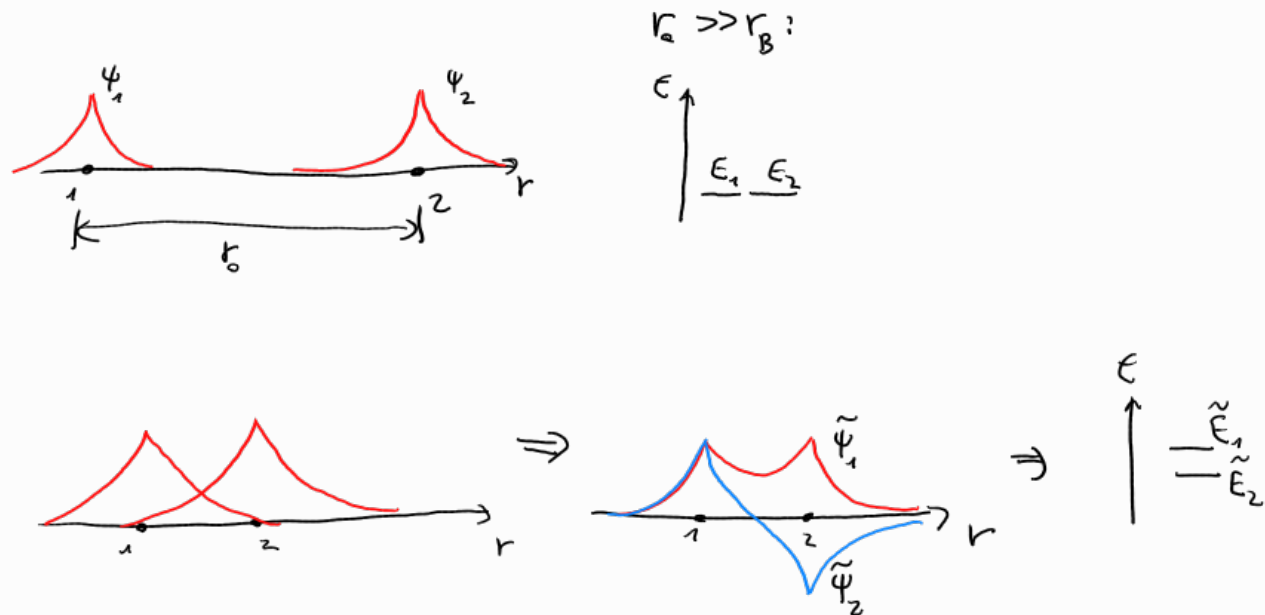
$\Rightarrow \langle \vec{p} \rangle = e_0 E_0 \hat{e}_x \tau \left(-\frac{\lambda \tau}{\lambda \tau - 1} e^{-t/\tau} + \frac{\tau}{\lambda \tau - 1} e^{-\lambda t} \right) =$
 $= e_0 E_0 \hat{e}_x \tau^2 \left(\frac{e^{-\lambda t} - \lambda e^{-t/\tau}}{\lambda \tau - 1} \right)$

$e = \int_0^\infty I dt = \int_0^\infty S \cdot j dt = \frac{n e_0}{m} S \int_0^\infty \langle \vec{p} \rangle dt = \frac{N e_0}{m l} \frac{e_0 E_0 \hat{e}_x \tau^2}{\lambda \tau - 1} \int_0^\infty (e^{-\lambda t} - \lambda e^{-t/\tau}) dt =$

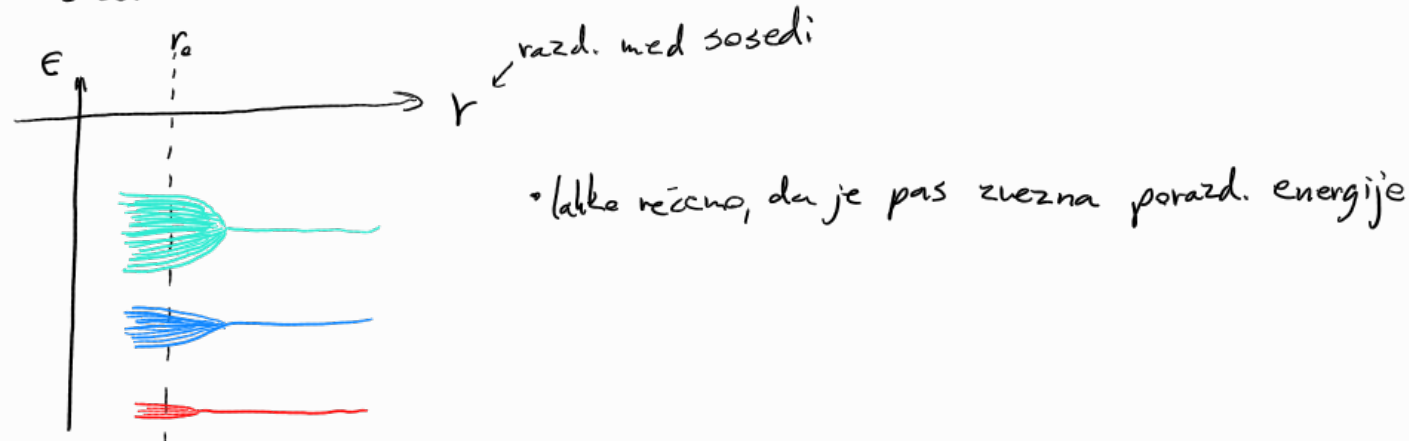
$= \frac{N e_0^2 E_0 \tau^2}{l m (\lambda \tau - 1)} \left(\frac{1}{\lambda} - \lambda \tau \right) = \frac{N e_0^2 E_0 \tau^2}{l m (\lambda \tau - 1)} \left(\frac{1 - \lambda^2 \tau}{\lambda} \right)$

Nastanek pasov

i) dvoatomna molekula (npr. H_2)



ii) kristal



Prost e^- vs. e^- v kristalu

$\psi(r) \propto e^{ikr}$ $\hookrightarrow$ Blochov teorem

$\hat{p} = -i\hbar \nabla$

$\hat{p}\psi \propto \hbar k \psi$

$\Rightarrow p = \hbar k$

$\psi_{\hbar k}(r) = u_{\hbar k}(r) e^{ikr} \longrightarrow \hat{p}\psi \neq \hbar k \psi$

Labels: indeks pasu, indeks e^- znotraj pasu, potencial zaradi jeder

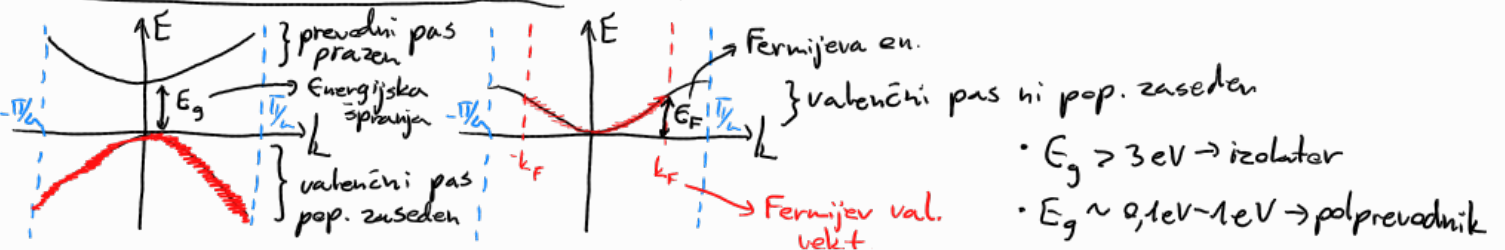
1. Brillouinova celica

$k \in [-\pi/a, \pi/a]$, a razd. med jedri

$k = \frac{2\pi}{Na} \ell \in \mathbb{Z} \Rightarrow \Delta k = \frac{2\pi}{Na}$

$\nearrow$ št. atomov

Izolator vs. kovina ($T=0$)



Čopić V/24
 • Dobra kovina ← e^- so prosti
 $E = 1 \text{ V/nm}$
 $\eta = 0,0032 \text{ m}^2/\text{Vs}$ → gibljivost, $\langle v \rangle = \eta E$
 $E_F = 7,0 \text{ eV}$

$\langle v \rangle = ?$
 $\frac{\langle v \rangle}{v_F} = ?$

$$E_F = \frac{\hbar^2 k_F^2}{2m} \quad \left. \begin{array}{l} \\ \hbar k_F = m v_F \end{array} \right\} v_F = \frac{\hbar k_F}{m} = \frac{\hbar}{m} \sqrt{\frac{2m E_F}{\hbar^2}} = 1,6 \cdot 10^6 \text{ m/s}$$

→ $\frac{\langle v \rangle}{v_F} = 2,0 \cdot 10^{-9}$

Čopić V/23

$\sigma_{Cu} = 5,9 \cdot 10^5 \Omega^{-1} \text{ m}^{-1}$
 $\sigma_{Na} = 2,2 \cdot 10^5 \Omega^{-1} \text{ m}^{-1}$
 $\rho_{Cu} = 8,9 \text{ g/cm}^3$
 $\rho_{Na} = 0,97 \text{ g/cm}^3$
 $Z_{Na} = Z_{Cu} = 1$
 $m_{Cu} = 63,55 \text{ u}$
 $m_{Na} = 22,99 \text{ u}$

$\langle l \rangle = v_F \cdot \tau \rightarrow \sigma = \frac{n e^2 \tau}{m} \rightarrow \tau = \frac{m \sigma}{n e^2}$
 $n(k_F) = ?$ zaradi spina
 $n = \frac{N}{V} = \frac{1}{V} \sum_{k=0}^{k_F} 2 = \frac{2}{V} \int \frac{d^3 k}{V_{k0}}$
 $\rho = \frac{m}{V} = \frac{N_{at} \cdot m_{at}}{V}$

3D kristal
 $k_0 = \frac{2\pi l}{Na} \rightarrow \vec{k}_{3D} = \frac{2\pi}{a} \left[\frac{l}{N_x}, \frac{m}{N_y}, \frac{n}{N_z} \right]$
 $\Delta k_{10} = \frac{2\pi}{Na} \rightarrow \Delta k_{3D} = \frac{(2\pi)^3}{V}$

$\tau_{Cu} = 2,5 \cdot 10^{-14} \text{ s}$
 $\tau_{Na} = 3,1 \cdot 10^{-14} \text{ s}$

$= \frac{2 \cdot V}{V \cdot (2\pi)^3} \int d^3 k = \frac{1}{4\pi^3} \int d\varphi d(\cos\theta) k^2 dk = \frac{4\pi}{4\pi^3} \cdot \frac{k_F^3}{3} = \frac{k_F^3}{3\pi^2}$
 $\downarrow \quad \downarrow \quad \downarrow$
 $[0, 2\pi] \quad [1, 1] \quad [0, k_F]$

→ $k_F = \sqrt[3]{3\pi^2 n} = \frac{m v_F}{\hbar} \rightarrow v_F = \begin{cases} 1,6 \cdot 10^6 \text{ m/s} ; Cu \\ 1,0 \cdot 10^6 \text{ m/s} ; Na \end{cases}$

→ $\begin{cases} \langle l_{Cu} \rangle = 40 \text{ nm} \\ \langle l_{Na} \rangle = 30 \text{ nm} \end{cases}$

Verjetnostna porazdelitev za prevodniške e^- v kovini

$\frac{dN}{dE} = \rho(E) \cdot f(E)$
 $\uparrow$ $\uparrow$
 število stanj na enoto energije $\cdot$ Fermi-Diracova porazdelitev
 $dg = \frac{4\pi k^2 dk}{(2\pi)^3} \cdot 2 = \frac{V}{\pi^2} k^2 dk$
 $\downarrow$ spin

$E = \frac{\hbar^2 k^2}{2m} \rightarrow \frac{dk}{dE} = \sqrt{\frac{m}{2\hbar^2 E}}$
 $f(E) = \frac{1}{e^{\beta(E-\mu)} + 1}$
 $\rho(E) = \frac{dg}{dE} = \frac{dg}{dk} \cdot \frac{dk}{dE} = \frac{k^2 V}{\pi^2} \sqrt{\frac{m}{2\hbar^2 E}} = \frac{VE 2m}{\hbar^2 \pi^2} \sqrt{\frac{m}{2\hbar^2 E}}$
 $\uparrow$ $\uparrow$
 št. stanj znotraj $[E, E+\delta E]$ $\uparrow$ št. stanj znotraj $[k, k+\delta k]$
 $= \frac{4\pi (2m)^{3/2} V}{h^3} \sqrt{E}$

$$\langle E \rangle = \frac{1}{N} \int_0^{\infty} \frac{dN}{dE} \cdot E dE \stackrel{T=0}{=} \frac{1}{N} \int_0^{E_F} g(E) E dE = \frac{3}{5} E_F$$

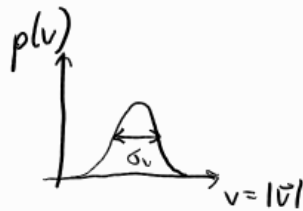
Zorko 4.6

$$Z=3$$

$$\rho = 2,79 \text{ g/cm}^3$$

$$M = 27 \text{ kg/mol}$$

$$T=0$$



$$f(E) = \begin{cases} 1; & E \leq E_F \\ 0; & E > E_F \end{cases}$$

$$\sigma_v = \sqrt{\langle v^2 \rangle - \langle v \rangle^2} = ?$$

$$E_F = \frac{\hbar^2 k_F^2}{2m}, \quad k_F = (3\pi^2 n)^{1/3}$$

$$\rightarrow g(E) = \frac{3}{2} N E_F^{-3/2} \sqrt{E}$$

$$\langle v \rangle = \frac{1}{N} \int_0^{E_F} g(E) v(E) dE = \frac{3}{2} E_F^{-3/2} \cdot \sqrt{\frac{2}{m}} \int_0^{E_F} E dE =$$

$$p = \hbar k = mv \rightarrow v = \sqrt{\frac{2E}{m}}$$

$$= \frac{3}{4} \sqrt{\frac{2E_F}{m}}$$

$$g = \frac{m}{V} = \frac{N_{\text{at}} \cdot M}{N_A \cdot V}$$

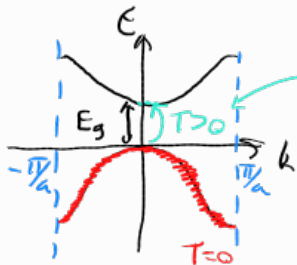
$$\langle v^2 \rangle = \frac{1}{N} \int_0^{E_F} g(E) v^2(E) dE = \frac{3}{m} \cdot \frac{3}{2} E_F^{-3/2} \int_0^{E_F} E^{3/2} dE = \frac{3}{m} \cdot \frac{2}{5} E_F = \frac{6}{5m} E_F$$

$$\sigma_v = \sqrt{\frac{6}{5} - \frac{9 \cdot 2}{16}} \frac{E_F}{m} = \sqrt{\frac{6 \cdot 8 - 9 \cdot 5}{40}} \frac{E_F}{m} = \sqrt{\frac{3}{40}} \frac{E_F}{m_e c^2} c$$

$$= 3,9 \cdot 10^5 \text{ m/s}$$

$$\left| \begin{aligned} h &= \frac{N_E}{V} = \frac{Z \cdot N_{\text{at}}}{V} \\ &= \frac{g N_A Z}{M} \end{aligned} \right| \quad E_F = \frac{\hbar^2 k_F^2}{2m}, \quad k_F = \sqrt[3]{3\pi^2 n}$$

Nedopirani (čisti) polprevodniki



pri dovolj veliki T bo nekaj e^- preskočilo v prevodni pas

- prevajanje bo posledica e^- v prevodnem pasu in vrzeli v valenčnem pasu
- $\exists$ simetrija med vrzelmi in e^- (vrzel bo na isti poziciji kot e^- , samo v drugem pasu)

$$E_F = \frac{\hbar^2 k^2}{2m}$$

• za okolice $k=0$ lahko vpeljemo m^* (efektivna masa), za katero velja $E = \frac{\hbar^2 k^2}{2m^*}$

Verj. porazd. za e^- in h^+ vrzeli

1) e^- v prevodnem pasu

$\rightarrow$ koliko e^- je v prevodnem pasu pri T?

$$\frac{dN_e}{dE} = g_e f(E)$$

$$\frac{4\pi(2m_e^*)^{3/2} V}{h^3} \sqrt{E - E_g}$$

minimum E je za E_g premanjšen nazgor

$$\frac{1}{e^{\beta(E-\mu)} + 1} \approx e^{-\beta(E-\mu)} = e^{-\beta(E_g-\mu)} e^{-\beta(E-E_g)}$$

$$\left. \begin{aligned} E - \mu &\sim 1 \text{ eV} \\ k_B T &\sim \frac{1}{40} \text{ eV} \end{aligned} \right\} e^{\beta(E-\mu)} \gg 1$$

$$n_e(T) = \frac{N_e}{V} = \frac{1}{V} \int_{E_g}^{\infty} \frac{dN_e}{dE} dE = \frac{4\pi(2m_e^*)^{3/2}}{h^3} e^{-\beta(E_g-\mu)} \int_{E_g}^{\infty} \sqrt{E-E_g} e^{-\beta(E-E_g)} dE$$

$$= \frac{4\pi(2m_e^*)^{3/2}}{h^3} e^{-\beta(E_g-\mu)} (k_B T)^{3/2} \Gamma(1/2) = \left[2 \left(\frac{2\pi m_e^* k_B T}{h^2} \right)^{3/2} e^{-\beta(E_g-\mu)} \right] = n_{0,e}(T) e^{-\beta(E_g-\mu)}$$

2) h^+ v valenčnem pasu

• analogno

$$\frac{dN_v}{dE} = g_v(E) \cdot f_v(E)$$

$$\frac{4\pi(2m_v^*)^{3/2}V}{h^3} \sqrt{-E} \quad 1 - f(E) = 1 - \frac{1}{1 + e^{\beta(E-\mu)}} = \frac{1}{1 + e^{-\beta(E-\mu)}} \approx e^{\beta(E-\mu)}$$

$$\rightarrow n_v = \frac{4\pi(2m_v^*)^{3/2}}{h^3} e^{-\beta\mu} \int_{-\infty}^0 \sqrt{-E} e^{\beta E} dE = \left[2 \left(\frac{2\pi m_v^* k_B T}{h^2} \right)^{3/2} e^{-\beta\mu} \right] = n_{g,v} e^{-\beta\mu}$$

$$\Rightarrow n_e n_v = n_g^2(T) e^{-\beta E_g}$$

$$n_0 = 2 \left(\frac{2\pi(m_e^* m_v^*) k_B T}{h^2} \right)^{3/2}$$

V čistem polprevodniku velja

$$n_e = n_v$$

Copy V/25

$$E_g = 0,72 \text{ eV}$$

$$m_e^* = 0,041 m_0 \quad 9,1 \cdot 10^{-31} \text{ kg}$$

$$m_v^* = 0,28 m_0$$

$$T_1 = 0 \text{ K}$$

$$T_2 = 300 \text{ K}$$

$$\mu(T_1) = ?$$

$$\mu(T_2) = ?$$

$$n_e(T_2) = n_v(T_2) = ?$$

$$n_e = n_v$$

$$n_{0,e}(T) e^{-\beta(E_g - \mu)} = n_{0,v}(T) e^{-\beta\mu}$$

$$m_e^{*3/2} e^{-\beta(E_g - \mu)} = m_v^{*3/2} e^{-\beta\mu}$$

$$\left(\frac{m_e^*}{m_v^*} \right)^{3/2} = e^{-\beta(E_g - 2\mu)}$$

$$\rightarrow \mu = \frac{1}{2} E_g + \frac{3}{4} k_B T \ln \left(\frac{m_v^*}{m_e^*} \right)$$

$$\mu(T_1) = \frac{1}{2} E_g = 0,36 \text{ eV}$$

$$\mu(T_2) = 0,397 \text{ eV}$$

$$n_e = n_v = n_{g,v} e^{-\beta\mu} = 7,75 \cdot 10^{17} \text{ m}^{-3}$$

Čopič V/27

$$E_g = 0,72 \text{ eV}$$

$$m_e^* = 0,41 m_0$$

$$\eta_e = 0,38 \frac{\text{m}^2}{\text{Vs}}$$

$$m_v^* = 0,28 m_0$$

$$\eta_v = 0,18 \frac{\text{m}^2}{\text{Vs}}$$

$$T = 300 \text{ K}$$

$$\sigma = ?$$

$$j = n e_0 \langle v \rangle \quad \leftarrow \text{kovine}$$

$$\text{za polprevodnike} \rightarrow j = n_e e_0 \langle v \rangle_e + n_v e_0 \langle v \rangle_v \quad \langle v \rangle = \eta E$$

$$j = \sigma E$$

$$\rightarrow \sigma = e_0 (n_e \eta_e + n_v \eta_v) \quad \text{iz prejšnje nal.}$$

Čopič V/29

$$E_g = 0,72 \text{ eV}$$

$$T_0 = 20^\circ \text{C} = 293 \text{ K}$$

$$\frac{\Delta R}{R} = 0,001$$

$$\frac{\Delta T}{T} = ?$$

okoliščini T_0

$$\Delta T_{\min} = ?$$

$$\frac{1}{\sigma}$$

$$R = \frac{l}{\sigma S}$$



$$\ln R = \ln l - \ln \sigma - \ln S \quad /d$$

$$\frac{dR}{R} = \frac{dl}{l} - \frac{d\sigma}{\sigma} - \frac{dS}{S}$$

$$\sigma = n_e e_0 \eta_e + n_v e_0 \eta_v$$

$$n_e = n_v = n_0(T) e^{-\beta \frac{E_g}{2}}$$

$$\sigma = e_0 (\eta_e + \eta_v) n_0 e^{-\beta \frac{E_g}{2}}$$

$$\frac{d\sigma}{\sigma} = \frac{d(e_0(\eta_e + \eta_v))}{e_0(\eta_e + \eta_v)} + \frac{dn_0}{n_0} - d(\beta \frac{E_g}{2})$$

$$\frac{dn_0}{n_0} = \frac{1}{n_0} \frac{dn_0}{dT} dT = \frac{3}{2} \frac{dT}{T}$$

$\uparrow$ $K T^{3/2}$ $\uparrow$ $K \cdot \frac{3}{2} T^{1/2}$

$$d(\beta \frac{E_g}{2}) = \frac{d(\beta \frac{E_g}{2})}{dT} dT = \frac{E_g}{2 k_B} (-1) \frac{1}{T^2} dT = -\frac{E_g}{2 k_B T^2} dT$$

$$\rightarrow \left| \frac{dR}{R} \right| = \left| \frac{d\sigma}{\sigma} \right| = \left(\frac{3}{2} + \frac{1}{2} E_g \beta \right) \frac{dT}{T} \xrightarrow{\text{majhna spr.}} \frac{\Delta T}{T} = \frac{\frac{\Delta R}{R}}{3 + E_g \beta} = \underline{\underline{6,3 \cdot 10^{-5}}}$$

$$\Delta T_{\min} = \frac{\Delta T}{T} \cdot T_0 = \underline{\underline{0,02 \text{ K}}}$$

Dopirani polprevodniki

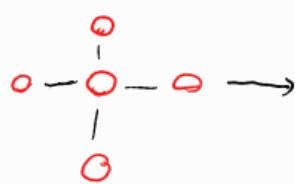
- v kristalni rešetki polprevodnika nečistoče (atomi nekega drugega elementa)
- predp. majhno dopiranje (npr. en atom / 10^6 atomov polprevodnika)

n-tip

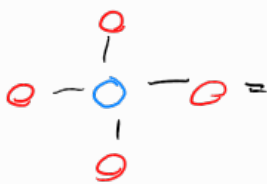
Ge + As

$^{32}_{32}\text{Ge}$

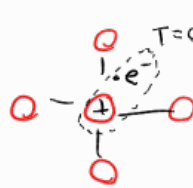
$^{33}_{33}\text{As}$



čisti Ge

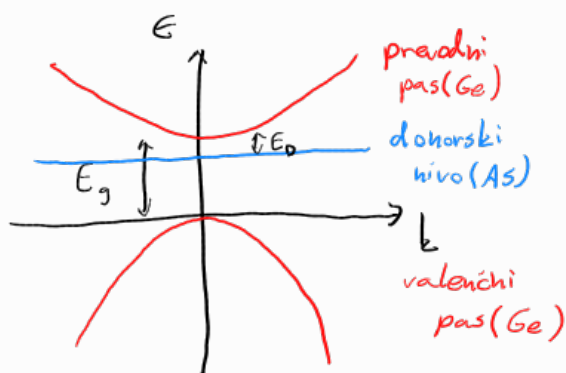


Dopiran



$T=0$: vezano stanje

$T>0$: e^- postane del kristala (preide v prevodni pas)



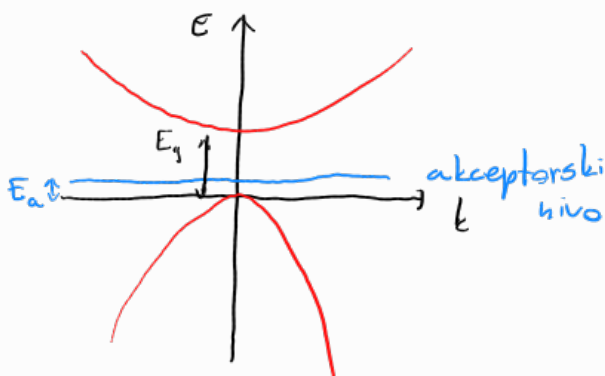
$T=0$: vsi e^- so v valenčnem pasu in don. nivoju

T nizka: večina e^- v prevodnem pasu pride iz don. nivoja

• prevajajo samo e^-

$$\underline{n_e \neq n_v}$$

p-tip



$T=0$: vsi e^- v valenčnem pasu

T nizka: e^- prehajajo na akceptorski nivo

• prevajajo samo vrzeli

$$n_e \neq n_v$$

• $n_e(T), n_v(T)$ sta enaka kot pri čistem polprevodniku (enačbe)

• μ se spremeni

• $n_e n_v = n_i^2(T) e^{-\beta E_g}$ še vedno velja

• $\Delta n = n_e - n_v = n_i \left[e^{\beta(\mu - \frac{1}{2}E_g)} - e^{-\beta(\mu - \frac{1}{2}E_g)} \right]$ (Strnad, za m^* dopirane je enaka m^* polprevodnika)

$$\rightarrow \mu = \frac{1}{2}E_g + \frac{1}{2}k_B T \frac{\Delta n}{n_i} + \dots$$

pri čistem polpr.

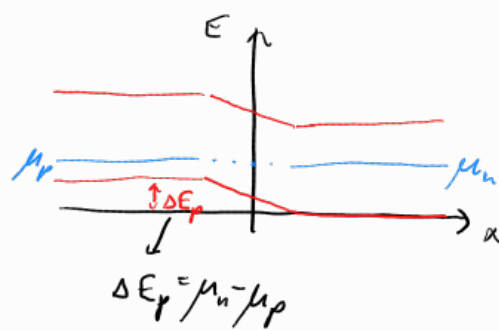
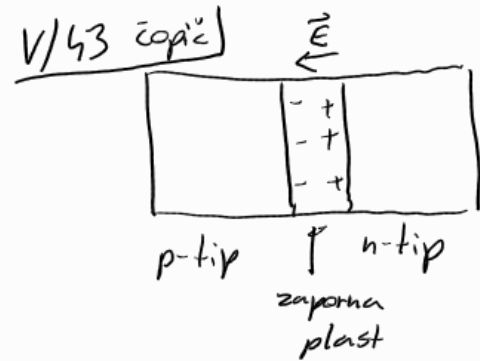
• $n_i = n_e = n_v \rightarrow$ lastno prevajanje

$n_1, n_0 = ? \rightarrow f_1, f_0 = ?$ (verj. za zasedenost akc./don. nivoja)

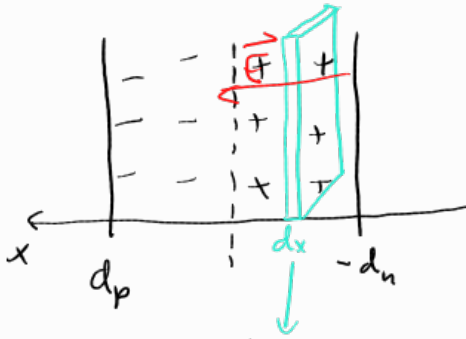
$$f_D(E_j) = \frac{1}{\frac{1}{2} e^{\beta(E_j - E_d - \mu)} + 1}$$

$$f_1(E_1) = \frac{1}{\frac{1}{2} e^{\beta(\mu - E_1)} + 1}$$

Manjka kvaje



Debelina zap. plasti:



Gauss: $\oint \vec{D} \cdot d\vec{S} = q$; posebej za d_n in d_p .

$$\vec{E}(x=-d_n)=0$$

$$\vec{E}(x=0)=\text{max}$$

$\vec{E} \cdot d\vec{S}$ bo ne ničeln samo na levi in desni plošči

$$\sum \epsilon_0 (E(x) + dE - E(x)) S = dq = \rho_e dV = \rho_e S dx$$

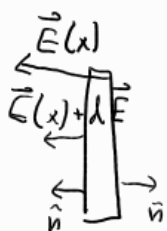
$$\sum \epsilon_0 dE = \rho_e dx$$

$\nwarrow n_e e_0 = n_0 e_0$

$$\rightarrow E(x) = \frac{n_0 e_0}{\sum \epsilon_0} x + k_1$$

$$E(x=-d_n)=0 \rightarrow k_1 = \frac{n_0 e_0}{\sum \epsilon_0} d_n$$

podobno za p-del:



$$\sum \epsilon_0 dE = \rho_e dx = -n_a e_0 dx$$

$$E(x) = -\frac{n_a e_0}{\sum \epsilon_0} x + k_2 \rightarrow E(x=d_p)=0 \rightarrow k_2 = \frac{n_a e_0}{\sum \epsilon_0} d_p$$

• zveznost v $E(x=0)$: $k_1 = k_2$ $n_0 d_n = n_a d_p$